Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Project AOS: Collaborative Development of an Agentic Operating System and On-Device VLA Models for Next-Generation Autonomous Robotics**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Digital Content Production and Distribution			
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	*Robot Control and Intelligence Technologies*		
Concept of Proposed Technology	**Core Concept** The proposed Agentic Operating System (AOS) is an AI-first, vertically integrated robot operating platform designed to overcome the fragmentation inherent in current autonomous robot system development, where the operating system, ROS2 middleware, AI models, and learning frameworks are loosely coupled in siloed architectures. Such fragmentation increases system complexity, limits performance optimization, and poses critical latency and security challenges, particularly for Agentic AI that requires real-time integration of large language models (LLMs), vision models, and robot control. AOS addresses these limitations by unifying low-level hardware control and high-level reasoning within a single coherent platform, enabling robots to operate as autonomous agents capable of perception, language grounding, planning, and action execution directly on edge devices. **Key Functions** 1) Open Web-Native Operating System (based on webOS OSE)			

	● Provides a proven, secure, and scalable OS foundation validated through large-scale commercial deployment. ● Adopts a service-oriented architecture (SOA) in which all system functions—from UI events to hardware control—are modularized as services and interconnected via the high-performance Luna Bus (LS2) IPC mechanism. ● Supports intelligent application lifecycle management through the System Application Manager (SAM), ensuring stability in resource-constrained edge robots. ● Enables secure remote software evolution through Firmware-Over-The-Air (FOTA) updates using A/B partitioning and delta updates. 2) Hybrid ROS 2 Middleware Integration ● Natively bridges ROS 2 with the webOS environment. ● A native C++ bridge for real-time, performance-critical functions such as motor control and SLAM. ● A JavaScript-based bridge for web UI integration, cloud connectivity, and high-level mission orchestration. ● Structurally combines ROS 2's robotic control capabilities with the productivity and extensibility of web technologies. 3) On-Device Agentic AI Core ● Integrates TensorRT-LLM and Isaac ROS on the NVIDIA Jetson Orin platform to execute Vision–Language–Action (VLA) models directly on-device. ● Enables robots to function as autonomous agents that understand natural language, perceive physical environments, plan actions, and use tools without reliance on cloud-based inference. **Potential Applications** 1) Autonomous service and inspection robots requiring low-latency reasoning and secure on-device intelligence 2) Industrial and logistics robots that must tightly integrate perception, planning, and control under real-time constraints 3) Robotic fleets benefiting from centralized lifecycle management and continuous over-the-air updates **4)** Next-generation AI-native robot platforms where agentic autonomy and scalable system integration are critical
Rationale for Joint R&D Project	**Critical Technical Barriers** Agentic AI is reshaping the paradigm of robotics; however, translating it into practical Physical AI systems faces two fundamental barriers. 1) Complexity of AI models: State-of-the-art Vision–Language–Action (VLA) models require massive computational resources, making optimization for edge devices essential.

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	2) Lack of system-level integration: Even advanced AI algorithms cannot move beyond laboratory prototypes without robust system software—such as a tightly integrated operating system and middleware—to support real-world deployment. **Why Global Collaboration?** This project aims to address critical technical challenges and lead global standards by integrating the complementary strengths of Korean and U.S. technologies, each representing best-in-class expertise in their respective domains. Possesses deep expertise in webOS Open Source Edition (OSE), embedded Linux, and hardware-level optimization. In particular, capability to build commercial-grade frameworks that integrate webOS and ROS 2, along with its FOTA-based platform lifecycle management technologies, represents a world-class system platform competence. Research institute holds a global reputation in Embodied AI, robot learning, and Vision–Language–Action (VLA) model research. It is especially recognized for its leading research outcomes in reasoning algorithms for physical interaction and Sim-to-Real transfer learning, which are critical for deploying AI in real-world robotic systems. By deploying Purdue University's state-of-the-art VLA algorithms directly onto AOS platform, the collaboration enables immediate validation of theoretical research outcomes at a commercially deployable system level. Through a co-design approach in which the operating system kernel and middleware are optimized in accordance with the requirements of AI models (Purdue), the project enables the realization of ultra-low-latency, high-efficiency robotic systems that cannot be achieved through isolated, component-level optimization alone.
Key Words	1) Agentic AI 2) Agentic Operating System (AOS) 3) Vision-Language-Action (VLA) Models 4) webOS OSE (Open Source Edition) 5) ROS 2 (Robot Operating System 2) 6) Federated Learning. 7) Physical AI

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3. Research and Development Objectives & Plan

Technology Readiness Level	Start of the Project	Level (3)	End of the Project	Level (7)

R&D Objectives

R&D Objectives

Beyond simple automation, this project develops an integrated operating system (AOS) and edge-based Vision–Language–Action (VLA) technologies for agentic robots that can understand context and make autonomous decisions. The overall system architecture is illustrated in Figure 1.

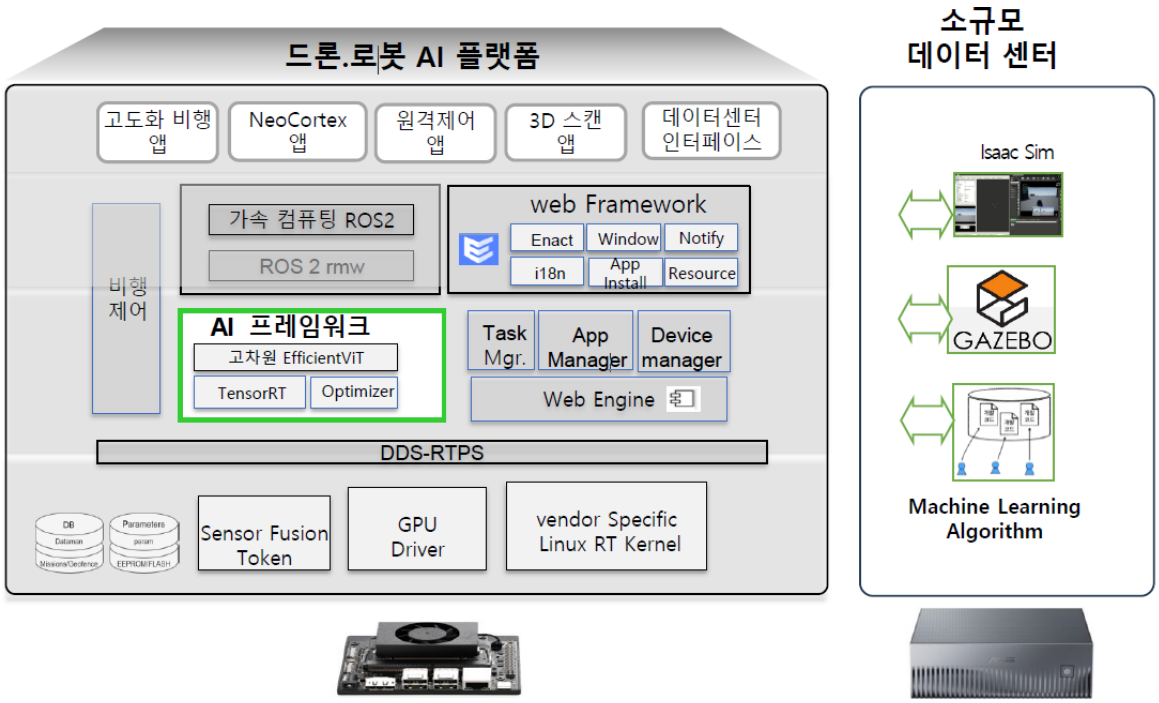

Figure 1. Vertical Integration Structure of AOS-VLA

Detailed Objectives

1) Vertical Integrated Service-Oriented Architecture (SOA): To address system fragmentation, the project integrates Open webOS (lightweight OS), ROS 2, and the NVIDIA AI stack into a single unified architecture.
   - Completion of the AOS platform through vertical integration of webOS OSE, ROS 2, and the NVIDIA AI stack.
   - Establishment of a high-performance yet flexible communication framework using a dual-bridge architecture (native C++ and JavaScript) to simultaneously satisfy real-time performance and developer productivity.
   - Luna Bus (LS2) functions as a central hub mediating communication among all system components—including UI, drivers, and AI modules—enabling unified system-wide monitoring and control from a single point.

2) On-Device VLA Optimization: Development of lightweight VLA models optimized for real-time inference on edge devices (e.g., NVIDIA Jetson Orin), together with TensorRT-LLM–based optimization.

- TensorRT-LLM: Accelerates LLM inference on NVIDIA GPUs through quantization (FP16 → INT8), kernel fusion, and in-flight batching, significantly reducing memory usage while dramatically improving throughput.
- Isaac ROS integration: Enables zero-copy data transfer using GPU acceleration, eliminating CPU overhead and improving perception and control performance.
- Maximization of grounding accuracy, converting natural language commands into precise robot control signals.

3) Self-Improving Learning Ecosystem: This project establishes a virtuous cycle of data collection and model updating.
   - Build a federated learning pipeline that leverages data collected from robot fleets, performing training locally on devices and transmitting only model updates (weights)—rather than raw data—to the server, thereby addressing privacy and bandwidth constraints.
   - Implement a continuous AI model deployment and update framework via webOS FOTA, enabling secure and efficient distribution of AI models and OS patches across the entire robot fleet using delta update mechanisms.

R&D Target Specifications

Specifiactions	Exsited	Target	Method
Natural language command success rate	< 60%	> 90%	Percentage of correctly executed actions in response to ambiguous natural language commands
Generalization Rate	< 40%	> 70%	Task success rate in unseen environments/objects (zero-shot performance)
Inference Latency	> 500ms	< 100ms	Input-to-action transformation latency of the VLA model running on edge devices, optimized using TensorRT.
Long-horizon Planning	〈30%	>80%	Completion rate of complex tasks consisting of sequential stages (decomposition → navigation → manipulation → return).
Transfer Efficiency	< 50%	>85%	Ratio of task performance retained in real physical environments relative to performance achieved during simulation-based training (e.g., Isaac Sim).

R&D Plan

Year 1: Core System Bring-up and Foundation Establishment

1) Port webOS OSE to the NVIDIA Jetson platform and optimize the kernel (applying PREEMPT_RT).
2) Develop an MVP of native C++ and JavaScript bridges for communication between webOS and ROS 2.
3) Establish a basic FOTA (OTA) update pipeline.
4) Design the baseline VLA model architecture for robot control.
5) Collect data and conduct initial model training in a simulation environment (Isaac Sim).

6) Design interface logic that converts natural language commands into ROS 2 messages.

Year 2: Intelligence Integration and Physical AI Implementation

1) Integrate the TensorRT-LLM runtime as an AOS system service.
2) Optimize interoperability between Isaac ROS (Perception/Nav2) stacks and webOS services.
3) Implement a shared-memory architecture for high-speed processing of sensor data (camera/LiDAR).
4) Develop and deploy lightweight on-device VLA models (with quantization applied) within AOS.
5) Implement Physical AI by developing models that predict the outcomes of physical interactions.
6) Advance Sim-to-Real transfer learning algorithms and initiate testing in real-world environments.

Year 3: Agentic Evolution and Ecosystem Expansion

1) Develop federated learning client modules and integrate them into AOS.
2) Complete and demonstrate a data flywheel through an automated loop of
3) [data collection → training → FOTA deployment].
4) Conduct field testing using multiple robots (fleet-level testing) and verify system stability.
5) Complete high-level agent logic capable of autonomous goal decomposition and tool use.
6) Develop reflection and self-correction algorithms for failure recovery.
7) Perform final performance evaluation in complex scenarios (e.g., infrastructure inspection).

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute is responsible for the core AI algorithms that constitute the “brain” of the robot in this project. 1) VLA Model Architecture: ● Design and train a multimodal model that fuses visual information and natural language commands to directly generate robot control actions. 2) Physical Reasoning: ● Develop high-level reasoning algorithms that go beyond simple pattern matching by understanding physical laws and causal relationships to predict the outcomes of actions. 3) Agentic Action Planning: ● Develop ReAct-based agent logic that decomposes complex and abstract goals (e.g., “clean the room”) into concrete sub-tasks and dynamically adapts plans according to situational context. 4) Sim-to-Real Transfer: ● Develop transfer learning techniques that effectively transfer intelligence learned in simulation environments to real robotic systems.
Expected R&R of Korean SME Partner	Korean SME partner is responsible for the system software that serves as the “body” and “nervous system” of the robot in this project. 1) AOS Architecture Development: ● Build an operating system based on webOS OSE that enhances real-time performance and natively integrates ROS 2 middleware to efficiently manage hardware resources. 2) Middleware Bridge Implementation: ● Develop a C++ bridge for high-performance processing and a JavaScript bridge for flexible application development, ensuring bottleneck-free data flow between AI models and robot hardware. 3) Edge AI Optimization: ● Integrate NVIDIA JetPack and TensorRT-LLM at the OS level to enable Purdue University’s VLA models to achieve maximum performance on edge devices. 4) FOTA and Fleet Management: ● Provide a platform for the secure deployment and management of trained models across multiple robots in a fleet.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Impact** 1) The first Agentic-dedicated operating system ● Secure the world's first agentic robot–specific operating system, optimized for AI execution from the OS kernel level, rather than simply running AI applications on top of a general-purpose OS. 2) A quantum in robotic intelligence: ● Through VLA-based agentic AI, robots achieve true autonomy by acting based on situational understanding and decision-making, rather than following predefined scripts. 3) Continuously evolving systems: ● Implement a self-improving system in which robots become increasingly intelligent as they operate, enabled by a data flywheel that combines federated learning and FOTA. **Economic and Industrial Impact** 1) Reduced development cost and time: ● By combining webOS's web-based development environment with the ROS 2 ecosystem, the platform lowers the entry barrier for robot application development and dramatically accelerates development cycles. 2) Creation of new service markets: ● Accelerate the commercialization of high-cognition robotic services across diverse domains, including smart factories, logistics, disaster response, and domestic service robots. 3) Global leadership in robot AOS platforms: ● Establish technological leadership in the global robotics market by presenting a successful international collaboration model that integrates Korea's strengths in manufacturing and system platforms with the United States' leadership in foundational AI technologies.

6. References

(1) "Motion Planning Transformers: One Model to Plan Them All" (2023)

(2) "Co-optimization of Robot Design and Controller using Reinforcement Learning"

(3) "Social-BiGAT: Multimodal Trajectory Forecasting using Bicycle-GAN and Graph Attention Networks" (NeurIPS)